Lesson 9

6 What is the value of $(3x-2)(4x+1)-(3x-2)4x+1$ when $x=4$?

(2002 AMC 10B Problems, Question #4)

A. 0 B. 1 C. 10 D. 11 E. 12

2 Factorize: $x^2(a+b)^2-2xy\left(a^2-b^2\right)+y^2(a-b)^2$

3 Factorize: $(p+q)^3-3(p+q)^2(p-q)+3(p+q)(p-q)^2-(p-q)^3$

HW

3 Factorize the following expressions:

(1) $2(x + y) + 6(x + y)^2 - 4(x + y)^3 =$ ______ .

(2) $(a + b)^3 (c + d) - (a + b)(c + d)^3 =$ ______ .

5 Factorize the following expressions:

(1) $9(m - n)^2 - 4(m + n)^2$

6 Factorize the following expressions:

(2) $\frac{1}{3}x^4y^4 + 9xy =$ ______ .

7 Factorize the following expressions:

(1) $9(x-3y)^3 + (3y-x) =$ ______ ;

(2) $\frac{1}{3}a^2 - 3b^2 =$ ______ ;

(4) $-20x^2 + 9x + 20 =$ ______ .

10 (Challenging) Given that $a = 20222020 \times 20222022^3 - 20222021 \times 20222019^3$, prove that a is a perfect cube, and write down the value of $\sqrt[3]{a}$.

L9 Extention

3 Factorize: $-3abx^4 + acx^3 - ax =$ ______ .

5 Factorize: $x^2 + 144y^2 - 25xy =$ ______ .

8 Factorize: $-2(y-x)^{2n}+4(x-y)^{2n-1}=$ _____ .

4 Factorize: $(2x-3y)(3x-2y)+(2y-3x)(2x+3y)-(6x-4y)(6x-9y)=$ _____ .

6 Factorize: $x^2-3xy+2y^2=$ _____ .

8. Factorize: $x^2 - (a + \frac{1}{a})x + 1 =$ ______ .

10. Factorize: $(a^2 - 6)^2 - 4a(a^2 - 6) - 5a^2 =$ ______ .